

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 20: Gas Turbines: Inlets & Compressors

Syllabus Source: CAE Oxford (pp. 139-194 (Pwr)) | Pass Target: >= 90% AviationExam

1. THE BRAYTON CYCLE & ENGINE INLETS

Gas turbine engines operate on the open Brayton cycle (continuous combustion at constant pressure). Intake ducts must deliver undisturbed, subsonic air uniformly to the first compressor stage at approximately Mach 0.4 to 0.5:

- Subsonic Air Inlets: CONVERGENT-DIVERGENT geometry on supersonic aircraft, but a simple DIVERGENT DUCT on subsonic airliners. As subsonic air flows through a divergent intake: Air velocity DECREASES while static pressure and temperature INCREASE (diffuser effect).
- Ram Recovery: At high forward airspeed, ram compression inside the intake restores total pressure above ambient atmospheric pressure, enhancing mass airflow and engine thrust.

2. COMPRESSOR TYPES: CENTRIFUGAL VS AXIAL FLOW

Compressors compress air prior to combustion to maximize thermal efficiency (Pressure Ratio = P_{exit} / P_{inlet}):

Feature	Centrifugal Flow Compressor	Axial Flow Compressor
Design Elements	Impeller (rotor), Diffuser (stator), Manifold.	Alternating rows of rotating Rotor Blades and stationary Stator Vanes.
Airflow Path	Air enters eye, accelerated radially outwards 90°.	Air flows parallel to engine centerline; cross-sectional annulus tapers.
Stage Pressure Ratio	High per single stage (~4:1 to 5:1).	Low per stage (~1.2:1), but cumulative multi-stage pressure ratio exceeds 30:1 to 45:1!
Frontal Area & Drag	Large frontal area; high aerodynamic drag.	Small frontal area; low drag; ideal for high-speed transonic flight.
FOD Resistance	High resistance to foreign object damage and ice.	Susceptible to blade damage, tip rubbing, and aerodynamic stalls.

3. COMPRESSOR STALL & COMPRESSOR SURGE

Compressor blades are miniature aerodynamic airfoils operating at an angle of attack (Alpha = Vector sum of rotational velocity and axial airflow):

- Compressor Stall: Localized aerodynamic stall of one or more rotor blades. Airflow separates from the suction side of the blade, forming rotating stall cells that propagate around the annulus in the direction opposite to rotation relative to the blades.
- Compressor Surge: Complete catastrophic breakdown of airflow across the entire compressor! High-pressure air from the combustion chamber rushes violently forward in reverse out the engine intake with loud bangs, violent airframe vibrations, rapid EGT spikes, and power loss!
- Anti-Stall / Anti-Surge Devices: 1. Multi-Spool Design (separates N1 Low Pressure and N2 High Pressure shafts so each spool runs at optimal aerodynamic speed); 2. Variable Stator Vanes (VSVs) and Variable Inlet Guide Vanes (VIGVs) that modulate stator vane angles; 3. Compressor Bleed Valves that dump excess intermediate stage air overboard at low RPM.

[!] EXAM TRAP (AVIATIONEXAM) - Axial Compressor Blade Angle of Attack Dynamics

- Blade Angle of Attack depends on: 1. Rotational Speed (RPM); 2. Axial Airflow Velocity (V_{axial}). - STALL RISK: If axial airflow DECREASES (e.g. crosswinds, inlet distortion) or if rotational RPM INCREASES rapidly, blade Angle of Attack INCREASES toward critical stall angle! - High stage compression causes compressor rear stages to be prone to choke at low RPM, while front stages are prone to stall!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Compressor Stall vs Surge Recovery**

Symptoms: Bangs, violent yaw/vibration, surging N1/N2 gauges, rapidly rising EGT. Recovery Procedure: Step 1: Retard affected engine thrust lever smoothly to IDLE. Step 2: Monitor EGT dropping to normal idle temperatures. Step 3: Bleed valves open and Variable Stator Vanes (VSV) adjust to relieve back-pressure. Step 4: Slowly advance thrust lever only if engine parameters stabilize.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: In an axial flow compressor of a gas turbine engine, what happens to the velocity and static pressure of the air as it flows through the compressor?**

- [A]: Velocity increases and static pressure decreases
- [B]: Velocity decreases (diffuser action) and static pressure increases
- [C]: Both velocity and static pressure increase
- [D]: Velocity remains constant while pressure drops

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The passages between stator vanes form divergent ducts. As air passes through, kinetic energy is converted into static pressure: air velocity decreases while static pressure increases.

Q2: What is the primary operational difference between a compressor stall and a compressor surge?

- [A]: A stall occurs on the turbine, a surge occurs on the fan
- [B]: A stall is a localized aerodynamic breakdown on individual compressor blades, while a surge is a complete reversal of airflow across the entire engine
- [C]: A surge occurs only when descending
- [D]: A stall increases engine thrust

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A stall affects isolated blades or rotating cells. Surge is the complete loss of pressure containment, where combustion chamber pressure forces air explosively backward out the intake.

Q3: What device prevents front-stage compressor stalling during engine acceleration or low RPM operation?

- [A]: Fuel jettison valves
- [B]: Variable Stator Vanes (VSVs) and compressor inter-stage bleed valves
- [C]: Thrust reversers
- [D]: Active clearance control

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At low RPM, rear stages restrict flow, backing up air and causing front stages to stall. Bleed valves open to vent air overboard, while VSVs tilt to maintain optimum blade angle of attack.